

SafeSight — Industrial PPE & Hazard Compliance Management System

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1) System Title + Domain

System Title

SafeSight — Industrial PPE & Hazard Compliance Management System

Application Domain

A computer vision–supported industrial safety system that detects PPE compliance and workplace hazards in real time and manages alerts and incidents through a structured workflow.

2) Problem Statement

Industrial environments (factories, warehouses, logistics areas) face recurring safety risks due to missing Personal Protective Equipment (PPE) and hazardous conditions such as fire, smoke, and unsafe forklift interactions. In many organizations, safety monitoring relies on manual observation, periodic inspections, or post-incident CCTV review. This approach is reactive and often results in delayed detection, incomplete documentation, and repeated violations.

Workers may unintentionally fail to wear required PPE such as helmet, vest, gloves, or goggles, while hazards like smoke or fire require immediate action to prevent injuries and operational disruption. Forklift presence in pedestrian zones is also a frequent source of serious accidents. When incidents are not captured and tracked consistently, safety officers and supervisors cannot reliably identify patterns, enforce corrective actions, or measure improvement.

SafeSight addresses this problem by using a YOLO11 object detection model (8 classes) as a real-time detection engine and by providing an incident management workflow for validating alerts, assigning corrective actions, and producing compliance reports.

YOLO11 Detection Classes (8):

0: helmet – 1:vest – 2:person -3:fire – 4:smoke – 5:gloves – 6:forklift – 7:goggles

3) Stakeholders and User Roles

3.1 Stakeholders

Workers:

Directly exposed to risks and affected by warnings, corrective actions, and safety outcomes.

Safety Officers (HSE Department):

Responsible for compliance monitoring, incident validation, documentation quality, and reporting.

Supervisors / Shift Managers:

Responsible for operational control, corrective action execution, and ensuring incidents are resolved.

Company Management:

Accountable for compliance results, accident reduction, and KPI-based safety performance improvement.

IT / System Administrators:

Responsible for system availability, configuration, user access control, and maintenance.

Regulatory Auditors (internal/external):

Require reliable incident records and compliance evidence for audits and reviews.

3.2 User Roles

Role A — Safety Officer (HSE) (Primary Compliance Authority)

Responsibilities

- Monitor and review system-generated alerts and incident evidence.
- Validate incidents (confirm/reject) and record notes and severity.
- Manage incident lifecycle (open → verified → resolved/false alarm) and generate compliance reports.

Role B — Supervisor / Shift Manager (Operational Authority)

Responsibilities

- Track incidents in assigned zones and coordinate corrective actions.
- Update incident status and resolution progress.
- Escalate recurring violations and support operational enforcement.

Role C — Worker (End User / Affected Individual)

Responsibilities

- Receive safety warnings and follow corrective instructions.
- Acknowledge alerts and improve PPE compliance behavior.
- Optionally provide feedback when an alert is incorrect.

Role D — System Administrator (Technical Role)

Responsibilities

- Configure zones, thresholds, and system parameters.
- Manage user accounts, roles, and access permissions.
- Monitor system health (camera connectivity, logging, storage status).

4) System Boundary (In-Scope / Out-of-Scope)

In-Scope

- Ingest live video streams from industrial cameras (RTSP/USB).
- Perform real-time detection using YOLO11 (8 classes).
- Detect PPE compliance and hazards (fire/smoke/forklift) and identify potential violations.
- Trigger alerts based on detection events and configurable thresholds.
- Create incident records automatically with timestamps, zone, type, and severity.
- Save evidence snapshots (and optionally short clips) for validated incidents.
- Provide a workflow for HSE validation (confirm/reject) and documentation.
- Allow supervisors to assign and track corrective actions and resolution status.
- Provide role-based authentication and access control.
- Produce summary reports and KPI-oriented compliance analytics.

Out-of-Scope

- Train, retrain, or re-label datasets for the YOLO model (model is treated as pre-trained).
- Facial recognition, identity verification, or employee-level biometric tracking.
- Full CCTV archival storage or forensic long-term video management.
- Physical device control (doors, sprinklers, alarms, access gates).
- HR payroll/penalty automation or disciplinary management systems.
- Enterprise-wide ERP/MES integration beyond basic export/reporting.
- Safety training LMS functionality.
- Replacement of human safety officers and supervisors.

High-Level Capabilities

- **Users can view real-time monitoring with detection overlays.**
- **The system can detect PPE elements (helmet, vest, gloves, goggles) and persons.**
- **The system can detect hazards (fire, smoke) and forklift presence.**
- **The system can generate real-time alerts with severity and zone information.**
- **The system can automatically log incidents and attach evidence snapshots.**
- **Safety Officers can validate incidents (confirm/reject) and add notes.**
- **Supervisors can assign corrective actions and update resolution status.**
- **The system can track recurring violations and compliance trends by time/zone/type.**
- **The system can generate weekly/monthly summary reports.**
- **The system can enforce role-based access and maintain basic audit logging.**

6) Mission / Vision

SafeSight's mission is to reduce industrial accidents by enabling proactive monitoring of PPE compliance and workplace hazards. By combining real-time computer vision detection with structured incident management, the system shortens response time and improves documentation quality. Its vision is to support measurable, KPI-driven safety improvement in industrial environments through scalable digital safety processes.

7) Objectives

- Reduce hazard detection-to-alert time to **under 2 seconds** under normal operating conditions.
- Reduce repeated PPE violations by **20% within 8 weeks** in monitored zones.
- Keep false alert rate below **10%** through validation and threshold tuning.
- Ensure at least **90% of confirmed incidents are resolved within 24 hours**.
- Reduce manual incident reporting effort by **50%** via automated logging and evidence capture.

8) Success Criteria / KPIs

- **Mean Time to Detect (MTTD):** average time from event appearance to alert generation.
- **False Alert Rate:** percentage of rejected alerts vs total alerts.
- **Incident Resolution Time:** time from confirmation to resolution/closure.
- **Compliance Trend:** change in PPE violation frequency over time.
- **Recurring Violation Index:** number of repeated incidents per zone/period.

9) Constraints + Risks

Constraint 1 — Camera quality and environmental conditions: low light, occlusion, and motion blur may reduce detection accuracy.

Mitigation: define minimum camera placement and lighting guidelines; tune thresholds per zone.

Constraint 2 — Privacy and retention policy requirements: storing evidence may be restricted.

Mitigation: store only incident snapshots/short clips with access control and a defined retention period.

Risk 1 — Excessive false positives may reduce user trust (alarm fatigue).

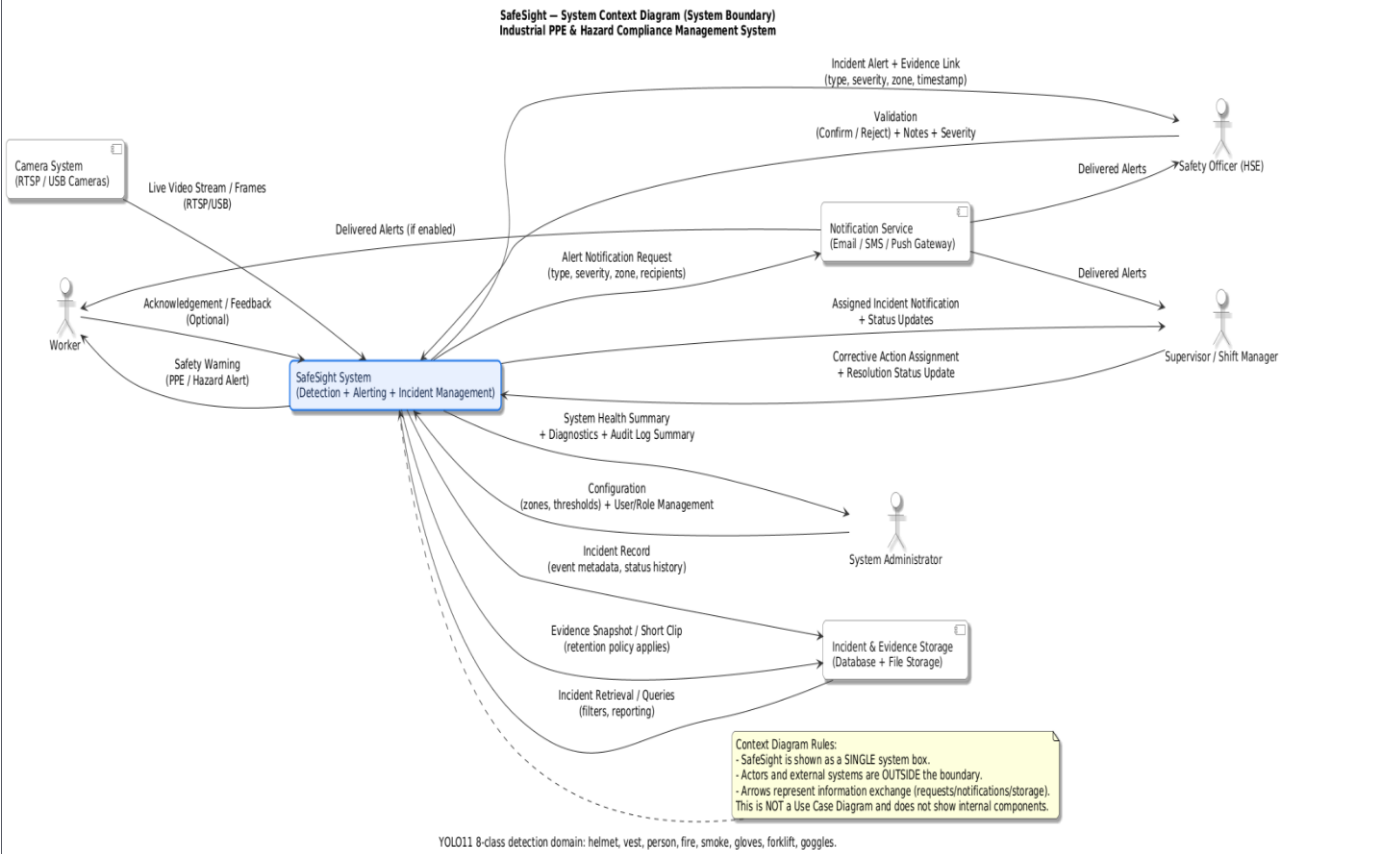
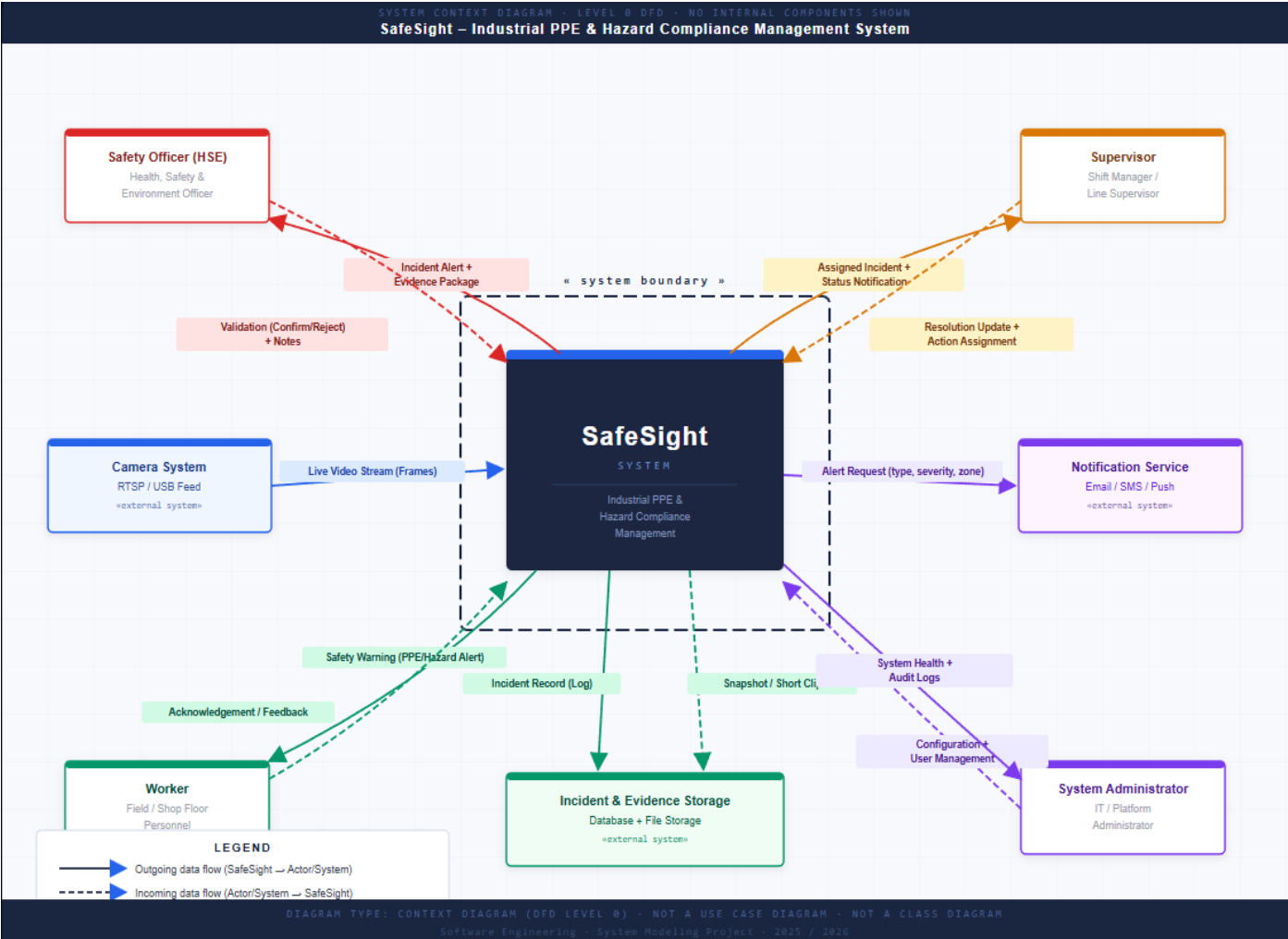
Mitigation: use confirmation workflow, time-threshold triggers, and zone-based configurations.

Risk 2 — Network or camera interruptions may stop live monitoring.

Mitigation: health checks, auto-reconnect, and "camera offline" notifications.

Risk 3 — Scope creep

Mitigation: keep boundary fixed to detection + alerts + incident management + reporting only.



10) REVIEW

The SafeSight system has been deliberately defined with a controlled and limited scope. The project focuses strictly on real-time detection using a pre-trained YOLO11 model, alert generation, incident validation, corrective action tracking, and reporting.

The system explicitly excludes model training, facial recognition, hardware control, ERP integration, and HR automation. These exclusions ensure that the system remains a focused detection and incident management platform rather than an enterprise-wide safety solution.

The In-Scope and Out-of-Scope sections are mutually exclusive and clearly defined.

10.2 Clarity and Consistency Review

Key terms are used consistently throughout the document:

- “Alert” refers to a real-time system notification.
- “Incident” refers to a recorded and managed safety event.
- User roles are clearly separated with non-overlapping responsibilities.
- Stakeholders are distinguished from system users.
- YOLO11 detection classes are consistently referenced.

All objectives are measurable and include quantitative targets.

All risks include mitigation strategies.

10.3 Final Validation

After reviewing system boundaries, terminology, responsibilities, and measurable goals, the team confirms that the SafeSight definition is clear, controlled, and ready to proceed to the requirements specification phase (D2).